Rotation Motion of Rigid Bodies: Moment of Inertia

SYNOPSIS

Notes 1 and 2 treated a *point* going round a circle. This Note changes the subject to a *body* turning about an axis, where the mass is spread out and the mass that sits far from the axis counts for more than the mass close to it.

Nothing here is asserted. Every rotational quantity comes out of the linear one you already have, by substituting the bridge relations — $\Delta d = r\Delta\theta$, $v_t = r\omega$, $a_t = r\alpha$. Momentum $p = mv_t$ becomes angular momentum $L = I\omega$; kinetic energy $\frac{1}{2}mv^2$ becomes rotational kinetic energy $\frac{1}{2}I\omega^2$; and $F = ma$ becomes $\tau = I\alpha$. In every one of them the same combination mr^2 falls out, and that combination is the moment of inertia — the **resistance to rotation**.

Then two things you can do with it. Angular momentum is conserved when nothing twists the body, which is why a skater speeds up when she pulls her arms in — and why her rotational kinetic energy does *not* stay the same, so somebody has to do that work. And a body that rolls without slipping has to put its initial potential energy into two places at once, so it arrives slower than one that slides.

BEFORE YOU READ ON — PREDICT

A hoop and a solid disc have the same mass and the same radius. Released together at the top of a ramp, they roll down without slipping. Write one sentence saying which reaches the bottom first, and why.

PERFORMANCE EXPECTATIONS

Use the bridge relations to convert between linear and angular quantities for a point on a rotating rigid body. Compute the moment of inertia of a point mass and of a standard shape, and use it to find angular momentum and rotational kinetic energy. Apply conservation of angular momentum. Apply $\tau = I\alpha$, the

angular impulse $\Delta L = \tau \Delta t$ and the rotational work $\Delta \text{RKE} = \tau \Delta \theta$. Set up the energy statement for a body rolling without slipping.

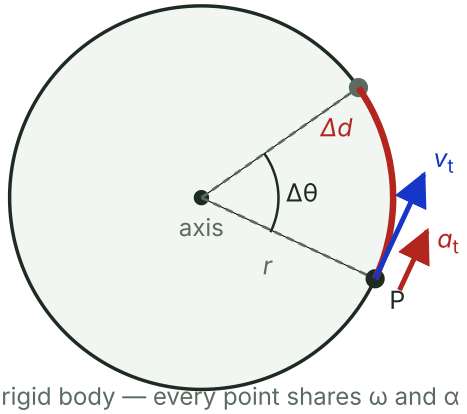
For PHY 151 — with calculus: Get the moment of inertia of a continuous body from $I = \int r^2 \, dm$ rather than from a table, and write Newton's second law for rotation in its general form $\boldsymbol{\tau} = d\boldsymbol{L}/dt$.

WATCH · THE NOTE WORKED THROUGH

Recording to come. Print the guided notes (link at the bottom of this page), watch, and fill them in as you go.

Moment of inertia

Start with the three relations that connect a point on a turning body to the body's own angular variables. They are the whole bridge, and every result in this Note is one of them substituted into something you already know.



one rigid body — every point shares ω and α
here the stone is speeding up, so a_t runs with v_t

the bridge

linear at radius r ←→ angular

$\Delta d = r\Delta\theta$	arc length
$v_t = r\omega$	tangential speed
$a_t = r\alpha$	tangential acceleration

θ in radians, always

The bridge. Each relation carries an r, which is why a point far from the axis moves further and faster than one close to it while both turn through the same angle.

1. A grindstone of radius 0.18 m is turning at 25 rad/s and speeding up at 4.0 rad/s². For a point on its rim: (A) what is the tangential speed, in m/s? (B) the tangential acceleration, in m/s²? (C) how far does that point travel along the rim while the stone turns through 2.0 rad?
2. A 0.50 kg ball is fixed to the end of a light 1.2 m rod that turns about its far end at 3.0 rad/s. Treat the ball as a point mass. (A) What is its moment of inertia about that end, in kg·m²? (B) Its

angular momentum, in kg·m²/s? (C) Its rotational kinetic energy, in J?

3. The same ball is slid to the middle of the rod, r = 0.60 m, and the rod is kept turning at the same 3.0 rad/s. By what factor does each of I, L and RKE change — and why is it the same factor for all three?
4. The grindstone in question 1 is a uniform disc of mass 12 kg, so $I = \frac{1}{2}Mr^2$. At 25 rad/s, what are (A) its moment of inertia, (B) its angular momentum, and (C) its rotational kinetic energy?

Show the working

Now do the two substitutions. They are the reason this Note exists.

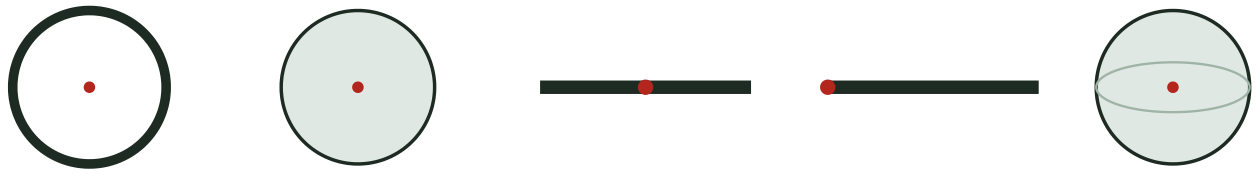
Angular momentum. Linear momentum of the point is $p = mv_t$. Its angular momentum about the axis is $L = rp$, so

$L = rp = rmv_t = rm(r\omega) = (mr^2)\omega \;\Rightarrow\; \boldsymbol{L = I\omega}$

Rotational kinetic energy. Kinetic energy of the point is $\frac{1}{2}mv_t^2$, and $v_t = r\omega$, so

$KE = \frac{1}{2}m(r\omega)^2 = \frac{1}{2}(mr^2)\omega^2 \;\Rightarrow\; \mathbf{RKE = \frac{1}{2}I\omega^2}$

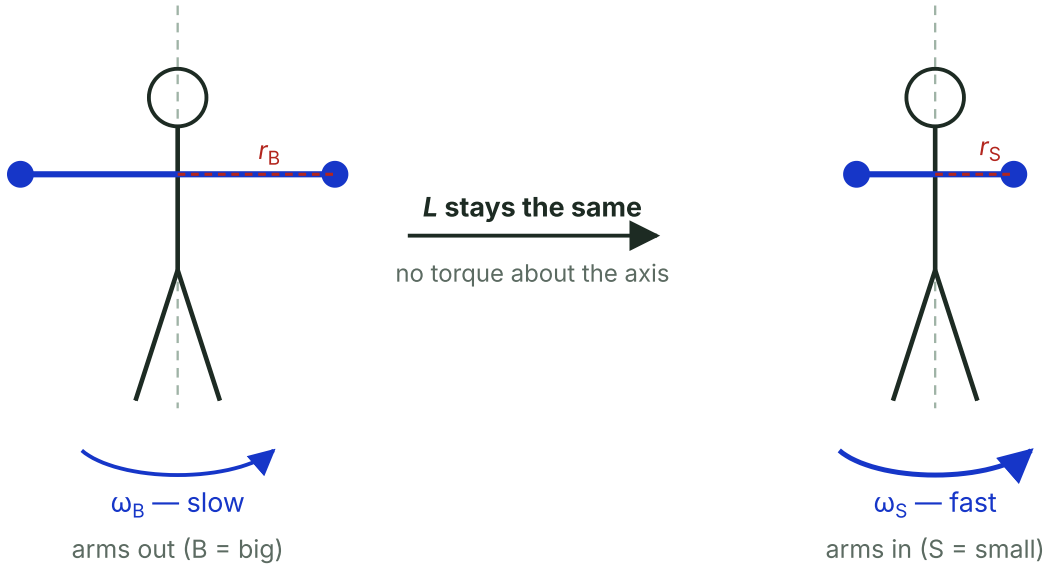
The same combination $\boldsymbol{I = mr^2}$ drops out of both. That is the **moment of inertia** — the resistance to rotation, standing in the same place that mass stands in $p = mv$ and $KE = \frac{1}{2}mv^2$. And L has a direction: for a body turning the way the figure above shows it, $\boldsymbol{L}$ points *into the page*. Curl the fingers of your right hand the way the body turns and your thumb points along $\boldsymbol{L}$.



The standard shapes. Red marks the axis. The *only* difference between the hoop and the disc of the same M and R is where the mass sits — all of the hoop's is at R, so its I is twice the disc's.

Angular momentum and rotational kinetic energy

If nothing twists a turning body — no net torque — its angular momentum cannot change. That single statement runs the skater, and it is worth being careful about what it does *not* say.



The skater. Only the arms move, and they move inward. *I* falls, so ω rises — but the rotational kinetic energy does not stay put, and the figure cannot tell you that. The algebra can.

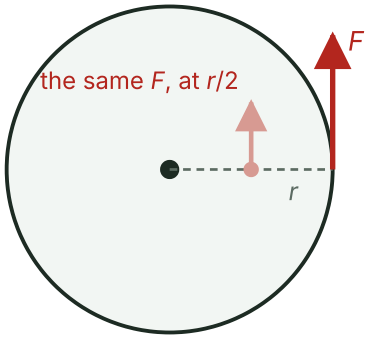
5. Treat the skater's arms as the only mass that moves, at radius r_B out and r_S in. Starting from $L_B = L_S$, show that $r_B^2\omega_B = r_S^2\omega_S$, and then that $r_B = 2r_S$ gives $\omega_S = 4\omega_B$. Which quantity is conserved here, and which is not?
6. She was turning at 1.2 rev/s with her arms out. With them pulled in to half the radius, what is her rate of turn — (A) in rev/s, and (B) in rad/s?
7. A student sits on a frictionless turntable holding two masses out at arm's length; the whole arrangement has $I_i = 6.0$ kg·m² and turns at 2.0 rad/s. He pulls the masses in until $I_f = 1.5$ kg·m². (A) What is ω_f? (B) What are the rotational kinetic energies before and after? (C) Where did the difference come from?

Show the working

Torque: Newton's second law for rotation

The last of the three substitutions, and the one that turns everything above into a way of predicting motion rather than just describing it.

force applied tangentially at the rim



$$\tau = rF = r(ma_t)$$
$$= r(m\,r\alpha) = (mr^2)\alpha$$
$$= \mathbf{I\alpha}$$

$\Delta p = F\Delta t \quad \leftrightarrow \quad \Delta L = \tau\Delta t$

$\Delta KE = F\Delta d \quad \leftrightarrow \quad \Delta RKE = \tau\Delta\theta$

half the radius, half the torque, half the angular acceleration

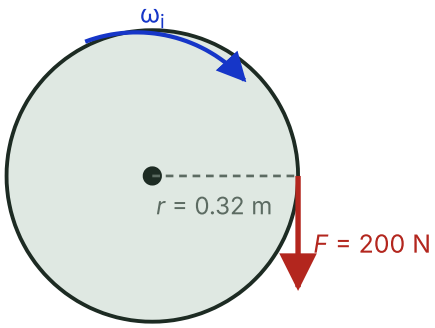
Substitute $a_t = r\alpha$ into $F = ma_t$ and multiply by r: mr^2 appears again, and Newton's second law comes out in angular form. The impulse and work statements follow the same way.

8. A wheel of moment of inertia 0.40 kg·m² is driven by a 25 N force applied tangentially at its rim, radius 0.30 m. (A) What torque does the force exert, in N·m? (B) What angular acceleration does it produce, in rad/s²?
9. That torque acts from rest for 3.0 s. (A) What is the change in angular momentum, from $\Delta L = \tau\Delta t$? (B) Find the final angular velocity two ways — from $L = I\omega$, and from $\omega = \alpha t$.
10. Instead, that torque acts through 4.0 rad from rest. (A) What is the rotational work done, from $\Delta RKE = \tau\Delta\theta$? (B) Find ω from $RKE = \frac{1}{2}I\omega^2$, and check it against $\omega^2 = 2\alpha\Delta\theta$.
11. The same 25 N force is applied at 0.15 m from the axis instead of at the 0.30 m rim. What happens to τ, to α, and to the angle the wheel turns through in the first 3.0 s?

Show the working

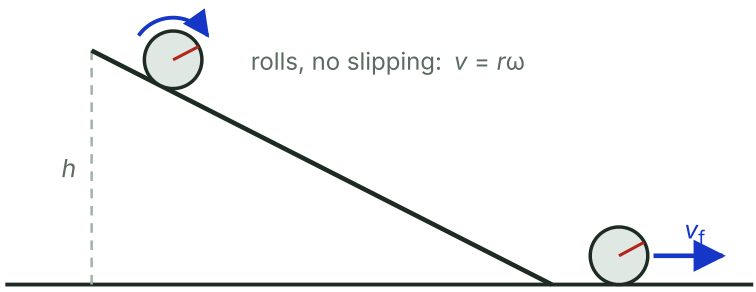
Roll without slipping

A body that rolls is doing two things at once: its centre is moving along, and it is turning. Rolling without slipping is the condition that ties them together, $v = r\omega$, and once it holds the energy has to be shared between the two.



M = 85 kg, $I = \tfrac{1}{2}Mr^2$

stops in $\Delta\theta$ = 1.0 rad



$$Mgh = \tfrac{1}{2}Mv_f^2 + \tfrac{1}{2}I\omega^2$$

the initial PE is partitioned into KE and RKE

Two uses of the same idea. On the left the rotational work equation stops a flywheel; on the right the energy statement has to feed two terms, so the marble arrives *slower* than one that simply slid.

12. A solid-disc flywheel of mass 85 kg and radius 0.32 m is brought to a stop by a 200 N friction force at its rim, acting through $\Delta\theta = 1.0$ rad. (A) How much rotational work was done on it, in J? (B) What is its moment of inertia, in kg·m²? (C) How fast was it turning before the brake was applied, in rad/s?

13. A marble — a solid sphere, $I = (2/5)Mr^2$ — rolls from rest without slipping down a slope of height h. Write the energy statement, substitute $\omega = v/r$, and solve for v_f. Compare it with the speed the marble would reach sliding down the same slope with no friction at all.

14. Take h = 1.5 m. (A) How fast is the marble going at the bottom, rolling? (B) How fast would it be going if it slid? (C) What fraction of the initial potential energy ended up as rotation?

Show the working

FOR PHY 151 — WITH CALCULUS

Two things the algebraic treatment has to take on trust: where the table of moments of inertia comes from, and why $\tau = I\alpha$ is the special case rather than the law.

15. For a continuous body $I = \int r^2\,dm$. Do that integral for a uniform disc of mass M and radius R about its axis, taking the ring at radius r of thickness dr, so that $dm = (M/\pi R^2)(2\pi r\,dr)$.

16. Write Newton's second law for rotation as $\boldsymbol{\tau} = d\boldsymbol{L}/dt$. What does that form give you that $\tau = I\alpha$ does not — and what does it say immediately about the skater?

Show the working

Looking ahead

With I, L, τ and RKE in hand, a rigid body turning about a fixed axis is fully described — and the lab that follows measures the one quantity of the four you cannot read off a ruler. **Note 4** is that lab's analysis: hang a mass from a string over a pulley, let it spin a disc, and verify $\tau = I\alpha$ by plotting α against τ and reading the moment of inertia off the slope.

Where this sits. Note 1 gave the angular variables, Note 2 the dynamics of a point on a circle, and this Note extended both to a body with its mass spread out. Every rotational equation you now have is a linear one with the bridge relations substituted in — there is nothing new to memorise, only the substitution to remember.

BACK TO YOUR PREDICTION

The **disc** gets there first. Both start with the same Mgh and both have to split it between $\frac{1}{2}Mv^2$ and $\frac{1}{2}I\omega^2$, but the hoop's $I = MR^2$ is twice the disc's $\frac{1}{2}MR^2$, so the hoop puts a bigger share into turning and has less left for getting down the ramp. Run the algebra of question 13 with each I and you get $v = \sqrt{(4gh/3)}$ for the disc against $\sqrt{(gh)}$ for the hoop. Neither answer contains M or R: it is the *shape* that decides, which is why a hoop loses to a disc of any size.

Check yourself

1. A thin hoop of mass 2.0 kg and radius 0.40 m turns about its centre. What is its moment of inertia, in kg·m²?

kg·m²

2. A skater spinning on frictionless ice pulls her arms in. Her angular momentum

- ○ increases, because she speeds up
- ○ stays the same, because nothing exerts a torque about her axis
- ○ decreases, because her moment of inertia decreases

- ○ stays the same, because her rotational kinetic energy does

3. In that same move, her rotational kinetic energy

- ○ stays the same, since no external force acts
- ○ decreases, since her moment of inertia decreases
- ○ increases, and the work is done by her own muscles
- ○ increases, and the energy comes from the ice

4. A net torque of 6.0 N·m acts on a body of moment of inertia 1.5 kg·m². What is its angular acceleration, in rad/s²?

_____ rad/s²

Check Clear

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

print or open on a tablet; record your prediction, then work the items as the video goes through them.

AFTER THE MODULE

Completed notes (PDF)

released when Module 5 closes.

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